

RobustUAVs.ai — v2

A Cross-Layer UAV Security Platform: Six Unified Corpora,
a Composed Network-to-Navigation Certificate, and a

Provenance-Enforcing Evaluation Surface

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Platform: <https://robustuavs.ai>

Artifact: <https://github.com/rogerpanel/RobustUAVs.ai>

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Abstract

RobustUAVs.ai is an open research platform for cross-layer UAV security. It exists to make one question answerable: *if an attacker stays just below a network detector's alarm threshold, will the mission still complete, and can that be proved?* Answering it requires objects no public dataset provides, so the platform supplies them.

The system spans **25 interconnected pages** in **8 sidebar groups**, served by a FastAPI control plane exposing **41 endpoints** across **16 modules**, with an Expo / React Native client of **27 screens** that runs from one source on web, iOS and Android. A registry carries **13 models** in five categories: four navigation detectors, two published baselines, one network detector, five certificates, and the composed guarantee that chains them.

Three properties distinguish it from a dashboard over a results directory. First, **nothing is hard-coded**: every number served traces to a file under `results/` or to a live call into the certificate engine, and a result that has not been produced returns HTTP 503 with the reason rather than an interpolated value. Second, **provenance is mandatory and visible**: the schema's unit of analysis is the cross-layer pairing, every pairing records how its causal coupling was established, and every page badges itself *measured* or *illustrative*. Third, the platform **reports against itself**: it states that the number of released `measured_same_platform` pairings is currently zero, and it says so on the page a reviewer would use to check.

The corpus unifies six sources totalling **431,773 events**, of which **200,610 (46.5%)** were captured on real hardware. The certificate engine implements four complementary methods and reports the fourth as a form awaiting its numeric constant rather than filling it. A live fleet simulation flies up to **12 aircraft** under **10 attack classes** with emergent mesh partitioning, and a six-phase GNSS spoofing process runs interactively. This document describes what is built, what is measured, what is modelled, and what is still blocked.

Version 2 adds two things around an unchanged core. A self-hosted **project mail** stack gives the platform four official addresses, with DKIM signing and Roundcube webmail, built alongside the web deployment without touching it. And the single roadmap of v1 has become a **research programme of five follow-on papers**, each of which names the platform page or engine extension it requires. v2 also corrects one statement in v1 about how the origin terminates TLS.

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1 Release Notes for Version 2

Area	Change since v1
Platform code	Unchanged. The 25 pages, 41 endpoints, 27 screens, 13 registered models and every committed result are byte-identical to v1; each v1 figure in this document was re-checked against source for this release.
Project mail	New (§12). Four mailboxes on <code>robustuavs.ai</code> , DKIM, Roundcube, Cloudflare DNS, and operator scripts. Built and verified offline; <i>not yet installed</i> on the server.
Deployment	Corrected (§11). The origin uses a Cloudflare Origin CA certificate with ACME disabled, not automatic TLS as v1 stated.
Research programme	Replaced (§16). v1's three follow-on directions are now five papers with named venues and the platform work each implies.
Runbook	<code>docs/deployment_runbook.md</code> gains a mail section and a cross-reference to the DNS trade-off mail introduces.

■ *Nothing in the platform's behaviour changed between v1 and v2, so no reproduced figure should move. The ten-minute check in §14 still ends with an empty `git diff results/`, and that remains the way to confirm it.*

2 Introduction

2.1 The problem the platform is built around

UAV security is studied from two directions that do not meet. Network intrusion detection inspects the traffic carried by the swarm mesh, the command-and-control link and the on-board bus, and raises an alarm when that traffic looks malicious. Certified robustness proves that a navigation model's output cannot move by more than a fixed amount when its sensor input is perturbed by no more than a given magnitude. The first stops at the alarm and says nothing about the flight that follows an intrusion it fails to catch. The second takes the perturbation as given and asks neither where it originates nor how often an adversary can produce one.

The gap between them is not academic. A detector must tolerate natural jitter to be usable, so an adversary who stays just under that tolerance is invisible by construction. What happens to the aircraft afterwards is precisely the question an operator, an insurer and a regulator need answered, and it is the question that falls between the two literatures.

2.2 What the platform contributes

RobustUAVs.ai is the instrument for closing that gap. It contributes four things, and the sections that follow describe each in turn.

1. A **two-layer schema** whose unit of analysis is the *cross-layer pairing*: an interval during which a network attack is active, joined to the interval of degraded navigation it induces, carrying a mandatory flag recording how strongly that causal link is evidenced (§4).
2. A **corpus** of six public sources unified under that schema by one lossless adapter each, with a detector-operating-curve harness that makes the detector's alarm threshold a swept parameter (§5).
3. A **certificate engine** implementing four complementary methods and the composition that chains a detector threshold to a mission-completion floor (§6).
4. A **web platform** that serves all of the above, adds live simulation, and enforces the provenance discipline in the user interface rather than only in the data (§3 onwards).

2.3 The one design rule

Every other decision in this system follows from a single rule:

■ *A number is shown with its provenance, or it is not shown. When a result has not been produced, the platform says which input is missing and where that gap is tracked. It never substitutes a plausible value.*

This is why a page can return HTTP 503, why the Calibration page deliberately displays nothing, why every screen carries a *measured* or *illustrative* badge, and why the corpus page reports a count of zero against the strongest evidence class. A reviewer who cannot tell a measurement from a model cannot check the work, and a platform that blurs the two is worse than no platform.

3 System Architecture

3.1 Two-plane architecture

The system separates a **research plane** that produces numbers from a **control plane** that serves them. The separation is strict and one-directional: the control plane reads committed artefacts and never computes a research result on the request path, except for the certificate engine, which is called live because it is deterministic and fast.

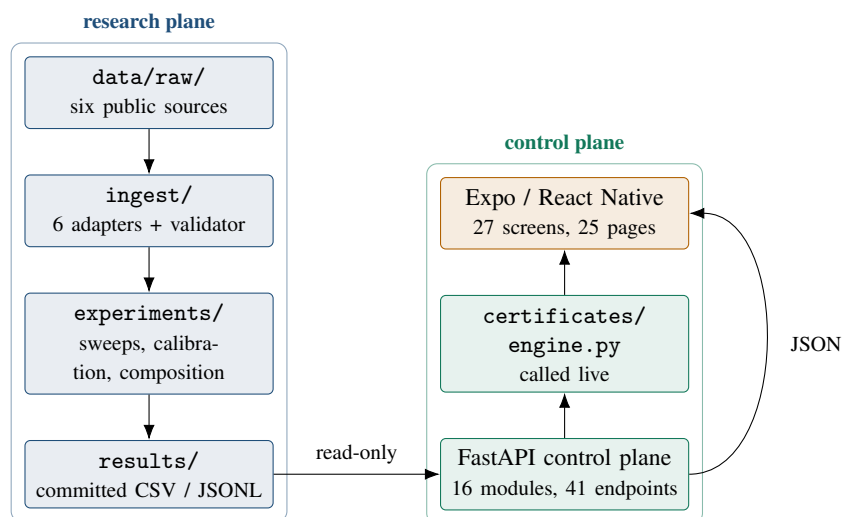


Figure 1: The two planes. Research artefacts flow one way. The control plane holds no research state of its own, which is what makes every served number traceable to a committed file.

3.2 Sidebar layout: 8 groups, 25 pages

Group	Pages	Contents
Results	2	Milestones, Composition
UAV / Aerial Defense	8	UAV Monitor, Swarm Graph, Fleet Demo, GNSS Spoof Monitor, Certification, Mission Plan, Perception, Dossier
Evaluation	5	Robustness, Ablations, ROC, Statistics, Calibration
Analyse	1	Upload & Analyse
Build	4	Agent Studio, Interface stability, Mission-distribution sensitivity, Federated Learning
Experiments	2	Runs, SOC Copilot
Corpus	2	Overview, Model registry
Present	1	Cover
Total	25	across 8 groups

Table 1: Navigation structure, read from `platform/mobile/src/layout/nav.js`.

Every page carries a hideable step-by-step help panel written for a reader who has not seen the page before, telling them what to look at first and what the page is evidence for. Several also carry a *tip* naming the reviewer question the page is designed to answer.

3.3 The 41 endpoints

Family	Endpoints
Health and registry	<code>/api/health</code> , <code>/api/models</code> , <code>/api/models/{id}</code>
Results	<code>/api/results</code> , <code>/api/results/{name}</code>
Certificates	<code>/api/certify/staleness</code>
Evaluation	<code>/api/eval/{robustness, ablations, roc, statistics, calibration, federated, interface, mission-distribution}</code>
UAV surface	<code>/api/uav/{overview, certificates, dossier, fleet, fleet/catalog, fleet/reset, fleet/step, gnss/run, gnss/run/reset, gnss/run/step, gnss/sky, swarm/snapshots, mission-plan/review, perception/catalog, ew-bench/curves, ew-bench/operating-point}</code>
Runs	<code>/api/runners</code> , <code>/api/runs (GET, POST)</code> , <code>/api/runs/{id}</code> , <code>/api/runs/{id}/cancel</code>
Copilot	<code>/api/copilot/{tools, tool/{name}, ask}</code>
Upload	<code>/api/upload/{analyse, compare, fly}</code>

Table 2: The control-plane surface. Interactive documentation is served at `/docs`.

3.4 Two response conventions that carry meaning

503 is not an error. A result endpoint returns 503 when the endpoint exists but its result file has not been produced in this deployment. The client renders this as “Not produced in this deployment” and points at `PENDING_ON_DATA.md`. Reporting it as a failure would misrepresent a data gap as a bug, and those are different things to a reviewer.

404 is a version mismatch. A 404 means the client is asking for something this API does not have, which in practice means the bundle is newer than the backend. The client says “Client and API are out of step” and gives the redeploy command. Conflating the two sends a reader to the wrong document.

4 The Two-Layer Schema

The schema exists to solve one problem: six datasets describe different things, at different rates, with no common clock, and any representation that forced them into one flat table would have to discard

whatever did not fit. It uses three record kinds, arranged in a hierarchy of increasing interpretation.

1. An **Event** is a single observation as its source recorded it, translated into common field names but never summarised. This layer is deliberately dumb: it asserts nothing beyond what the source file said. A `native` block preserves every unmapped source column, which is what makes ingestion provably lossless rather than merely convenient.
2. An **AttackWindow** is an interval during which one named attack was active on one layer. This layer asserts that something adversarial was happening, and when.
3. A **CrossLayerPairing** joins one network window to one autonomy window and asserts that the first *caused* the second. This is the only place in the schema where a causal claim is made, and therefore the only place carrying a mandatory evidence flag.

4.1 The `pairing_basis` flag

Value	Meaning
<code>measured_same_platform</code>	Both windows recorded on one testbed. The strongest evidence.
<code>physics_model</code>	Coupled through a named, versioned physical relation a reviewer can inspect and challenge.
<code>synthetic_alignment</code>	Aligned by construction, under a stated affine clock map with a bounded error budget.

A reviewer can filter the benchmark by evidence strength rather than trusting an unaudited join, and a claim can never silently upgrade its own provenance.

4.2 Three design choices

Time is always *relative* to each source's capture start, because the sources share no global clock, and cross-layer alignment happens at the window level, so a mismatch between a 50 Hz telemetry stream and a per-flow network record never forces a lossy resample. Attack labels are *unified* into a common taxonomy while the verbatim source label is retained in `label_native`. And every unmapped column is preserved.

Alignment therefore needs a stated rule and a stated error budget, since this is the step at which a benchmark could silently manufacture, or destroy, the effect it reports. Attack onsets are anchored under an affine map between the two relative clocks, with residual misalignment bounded by $e_{\text{align}} \leq 0.53$ s for every released pairing. Crucially, the quantity the certificate consumes is computed entirely within the network source, so misalignment can attach a pairing to the wrong autonomy window but cannot inflate or deflate the certified quantity itself.

5 The Corpus and Its Adapters

Source	Acquisition	Layer	Events	Role
HCRL UAVCAN	real testbed	network, bus	200,000	Bus flooding, fuzzing, replay
UAVIDS-2025	simulation (NS-3)	network, mesh	122,171	Large-scale mesh flows
UAV-EW-Bench-2026	simulation (physics)	autonomy	93,600	The navigation anchor
DATAMUt replay	simulation (event)	network, mesh	15,392	The sweepable detector
UAV Attack Dataset	real testbed	both	610	The only source observing both layers on one airframe
UAV-CAS	simulation (twin)	network, mesh	0	Adapter ready; release file not reachable
Total			431,773	200,610 real (46.5%)

Table 3: Corpus composition, regenerated by `experiments/provenance_ledger.py` so the document cannot drift from the artifact.

5.1 The number the platform reports against itself

Acquisition provenance is the weaker question. The stronger one is how many *pairings* rest on measurement, and there the honest answer is worse:

■ *The number of released `measured_same_platform` pairings is currently **zero**. The UAV Attack Dataset release does not carry machine-readable attack on/off timestamps, so the adapter cannot emit `AttackWindow` boundaries for it and refuses to infer them. Its three flights still supply the delay-to-position rate the certificate consumes, self-referenced to each flight’s own pre-attack median, so the headline result does not depend on closing this gap. But a reader who filters the artifact to the strongest evidence class will correctly retrieve nothing, and the platform says so on the Corpus page.*

5.2 Adapters and the validator

One adapter per source lives in `ingest/`, and `ingest/validate.py` is the gate: nothing enters `results/` unvalidated. `experiments/run_real_corpus.sh` runs the whole pipeline and is idempotent; a stage whose inputs are absent is skipped with a message rather than substituted.

■ *A caution that cost us a result. The third-party detector demo appends to its summary CSV. An early sweep harness therefore averaged every previous operating point into the current one, and the symptom was recall increasing with the threshold, which a threshold detector cannot exhibit. `experiments/sweep_full.py` now uses a fresh working directory per point. Anyone reimplementing the sweep should do the same.*

6 Models and Certificates

6.1 The registry: 13 models in five categories

Category	Count	Members
navigation	4	m1_ct_tgnn, m4_mambashield, m6_uc_hgp, m7_fedgtd
baseline	2	caf_cnn, seq2seq_tr
network	1	datamut
certificate	5	cert_gronwall, cert_smoothing, cert_staleness, cert_mwu, cert_pacbayes
composition	1	composed

Each card carries a provenance and a `certificate_status` field, a `constants` block, and a `caveat`. The caveat is not decoration: it is where a model states what it cannot do.

6.2 The composition

The chain the platform serves is

$$\theta \mapsto \Delta(\theta) \mapsto \delta(\theta) \mapsto \text{MCR} \geq f(\delta(\theta)),$$

read left to right: a detector’s alarm threshold θ bounds the residual undetected attack $\Delta(\theta)$; the residual maps to a bounded perturbation $\delta(\theta)$ on the navigation input; and a Lipschitz–Grönwall trajectory-tube argument turns that bounded input into a guaranteed Mission-Completion-Rate floor.

 *The chain is not a theorem end to end. Its first and last links are proved; the middle link, $\Delta \mapsto \delta$, is an empirical calibration. The platform and the accompanying manuscript both describe the result as a conditional guarantee under an empirically calibrated interface, and not as end-to-end certified security.*

6.3 Four certificates, and why one is not enough

Certificate	Bounds	Regime it covers
Lipschitz–Grönwall	the trajectory	Deterministic, bounded disturbance. The only one giving a per-instant geometric envelope, so the only one that discharges the corridor test.
Randomized smoothing	a decision at a point	Where the budget is known only statistically.
PAC-Bayes	the model	The gap between the training mission mix and this sortie.
MWU regret	the time axis	Cumulative loss against an <i>adapting</i> jammer. Vacuous at any single instant.

Because a trajectory, a decision, a model and a policy sequence are distinct objects, the four compose by conjunction rather than selection: a deployment reads its floor as the pointwise minimum, and the binding term identifies which assumption is closest to failing.

6.4 Instantiated constants

Constant	Value	Provenance
Lipschitz, measured local	$L = 1.181$	measured on visited trajectories
Lipschitz, global power iteration	$\hat{L} = 0.983$	<i>not used</i> : smaller, so it would flatter every bound
Grönwall radius at $T_c = 1$ s	0.1535	derived
Randomized-smoothing radius	0.44 at $\sigma = 0.25$	measured on the trained checkpoint
MWU policy set	$ S = 4$	stated
PAC-Bayes KL	pending	flagged, not substituted
Interface rate γ (supremum)	1.625 m/s	674 post-onset samples, real flights
Residual budget saturation	≤ 4.84 s	15,392 observed hops

The Lipschitz row is worth dwelling on. The local constant is the *larger* of the two, so it widens the trajectory tube and narrows the certified window. Quoting the global one would flatter every bound reported, which is why the platform uses the local value and says why.

7 The Interface Characterisation Campaign

The composition consumes γ , the metres of position error accrued per second of degraded navigation input, and the platform’s most consequential empirical question is whether γ is a single number. It is not.

7.1 The estimator decides the answer

Two estimators are available and they disagree qualitatively. Writing $e(t)$ for position error and t_0 for attack onset,

$$\gamma_{\text{cum}}(t) = \frac{e(t)}{t - t_0}, \quad \gamma_{\Delta}(t) = \frac{e(t) - e(t_0)}{t - t_0}.$$

Onset is declared when the error crosses a detection threshold, so $e(t_0)$ is not zero: it is 0.64 m for jamming at the receiver and 9.27 m for spoofing. Dividing that pre-existing offset by a small $t - t_0$ makes γ_{cum} diverge, reaching 8.6 m/s below one second.

■ *That number is an artifact of the detector, not a property of the attack, and reporting it would have been a serious error: 8.6 m/s exceeds $\gamma_{\text{req}} = 6.14$ m/s and would have declared the framework unusable on the strength of a measurement artifact. The delay budget is responsible only for error the attacker causes, so γ_{Δ} is the estimator the certificate is entitled to. Both are kept in the per-sample output so the contrast can be audited.*

7.2 What γ depends on

Over 674 post-onset samples, γ_{Δ} spans $6.1\times$ from median to supremum and varies systematically:

Factor	Effect	Mechanism
Attack type	$2.6\times$, spoofing $>$ jamming	Jamming removes information and the estimator dead-reckons; spoofing injects information the estimator accepts, so error accrues at the adversary’s chosen rate.
Satellite visibility	rises as satellites fall $14 \rightarrow 4$	Degraded geometry, weaker correction.
Staleness Error signal	peaks at $\tau \in [5, 10]$ s only $1.14\times$	Error saturates while the denominator keeps growing. The filter attenuates injected error far less than one might hope.

A descriptive least-squares fit on standardised covariates explains $R^2 = 0.79$, with physically sensible signs for satellite count (-0.22), staleness (-0.26) and airspeed ($+0.12$). No p -values are reported: samples within a flight are strongly autocorrelated and there are two attack flights, so any nominal significance would be an artifact of the dependence structure.

7.3 Coverage, stated as a limit

Factor	Status	Detail
Attack type	varied	jamming and spoofing
GNSS condition	varied, wide	satellites $14 \rightarrow 4$, HDOP $0.71 \rightarrow 4.16$, fix $3 \rightarrow 0$
Staleness	varied	0.4 to ~ 70 s
Airspeed	poor	hover; p90 of $\ v_{xy}\ < 0.65$ m/s
Platform	not varied	one PX4 / Pixhawk 4 / Holybro S500
Flight mode	not varied	hover / loiter only

A factor the release could not vary is reported as unvaried, never as showing no effect. The $6.1\times$ spread is therefore a *lower bound* on the variability a deployment would meet, and the two red rows need flights rather than analysis.

7.4 Mission-distribution sensitivity

MCR is a probability over a distribution of missions, so a bound on it describes a population and not an individual flight. The platform measures what that costs. Across nine mission distributions (three profiles $\times$ three receiver models) at a fixed defence:

- Marginalised over the whole 0–40 dB jamming sweep, the spread is **0.011**, which looks negligible.
- Conditional on attack strength, the mission profile alone moves MCR by **0.214**, and the receiver model by **0.154**.
- The jamming power at which a distribution first falls below the 0.90 reference ranges from 18.1 to 27.1 dB, a spread of **9 dB**, or roughly a factor of eight in adversary transmit power.

Averaging over the adversary hides a dependence that an operator, who flies one profile against one adversary, lives at.

8 The UAV / Aerial Defense Surface

8.1 Live fleet simulation

Up to **12 aircraft** fly a mission of nominal duration $T = 70$ s against a deadline of $(1 + \kappa)T$ with $\kappa = 0.25$. The dashboard shows exactly the number selected, and the fleet is instantiated by sampling from the five reachable corpus sources rather than from a random generator.

Class	Family	Effect
none	—	Nominal flight
fgsm	gradient	Single-step gradient sign; cheapest and weakest
pgd	gradient	Iterated with projection; the strong baseline
cw	optimisation	Carlini–Wagner
deepfool	geometric	Minimal-perturbation boundary search
gaussian	noise	Non-adversarial control
gnss_spoof	RF	Injects a false but consistent fix
jam_link	RF	Degrades mesh link quality
delay_relay	network	The class the theorem covers
label_flip	poisoning	Training-time

Table 4: Ten attack classes, from `platform/backend/app/uav.py`.

Mesh degradation is emergent rather than scripted: link residual is the sum of a measured baseline and accumulated staleness, and a link becomes delayed above 0.25 s of residual, and jammed above 5.0 s or when the worse endpoint drops below 25 % link quality. Jamming two of eight aircraft for 30 s partitions a nine-link mesh into four clusters with two isolated, which nothing in the code instructs it to do.

■ **Modelled versus measured.** The link residuals are sampled from real corpus data; the damage rates, return-to-launch thresholds and battery model are modelled. `docs/simulation_architecture.md` lists the four steps that would move the modelled quantities to measured, the shortest being per-step detector inference over the already-real link residuals. Every page badges itself accordingly.

8.2 GNSS spoof monitor

A six-phase interactive process rather than a static diagram: the user steps a spoofing attack through its phases, watches the satellite geometry and the estimator response, and can attach a *remark* at any step. The composed certificate is evaluated live at each phase.

8.3 The rest of the group

Swarm Graph renders mesh snapshots with a composite view at t_4 showing all selected aircraft together. **Certification** reads the locked certificate constants. **Mission Plan** reviews a proposed corridor and schedule against the certified window. **Perception** and **Dossier** present the model catalogue and the per-aircraft record.

9 Evaluation, Analysis and Copilot

9.1 Evaluation group

Robustness plots the certified floor against the detector threshold, one series per interface mapping, now including the campaign supremum at $\gamma = 1.625$ m/s. **Ablations** reports how tight γ must be for a given corridor to certify. **ROC** gives recall and false-positive rate against the threshold, and the generalisation to any scored detector. **Statistics** carries Wilcoxon signed-rank tests with Holm correction.

Calibration deliberately shows nothing. Expected Calibration Error needs per-hop detector *scores*, and the detector emits a boolean, so no reliability diagram is derivable from anything committed. The page returns `available: false` with the method, the exact missing input and what would unblock it. A reviewer asking “is your detector calibrated?” gets a straight answer about why the question cannot yet be answered, which is worth more than a curve fitted to a boolean.

9.2 Upload & Analyse

A reviewer can upload their own data, up to **1 GB** per file across **four** slots, and run the platform's models over it. Parsing is streaming with a reservoir sample of 4,000 rows for charts, so memory is bounded regardless of file size. Multi-dataset comparison supports leave-one-out cross-validation over the delay p95, an attack-transfer matrix, and a distribution-shift measure.

9.3 SOC Copilot

Four LLM providers in fallback order (anthropic, openai, gemini, deepseek) with a deterministic synthetic responder last, so the page never dead-ends. A caller-supplied key is used for one request and never stored on the server or in browser storage. Clickable thumbnails fetch results from pages the user has visited, so the copilot answers from what the reviewer has actually seen.

A smoke test enforces the load-bearing invariant: a tool advertised to the model but absent from the dispatcher is a lie told to the model, and CI fails on it.

10 Security, Sessions and Access

10.1 Per-user isolation without accounts

There is deliberately **no login or registration**, because the work is under double-blind review and reviewers must not be asked to identify themselves. Isolation is achieved with an `X-Session-Id` header generated per browser, which scopes the fleet, the run list and the rate allowance.

 *The session identifier is not a credential and the code says so where it is defined. It prevents two reviewers in different places from disturbing each other's simulation state. It is not an authentication mechanism and must not be treated as one.*

Rate limits are applied per session: 30 LLM calls and 20 uploads per hour.

10.2 Ethics gate

A four-section *Ethical Use Guidance and Caution* page gates the entire application, requiring two explicit acknowledgements before access. Acceptance is recorded locally against a version string, so revising the guidance re-prompts every user.

10.3 Deployment hardening

Cloudflare proxies every web hostname in *Full (strict)* mode. At the origin, Caddy terminates TLS with a Cloudflare Origin CA certificate — trusted by Cloudflare and by nothing else — and serves the static bundle; the API runs under systemd as `robustuavs-api`, bound to loopback. Ports 80 and 443 are meant to accept connections from Cloudflare's published ranges only, so the origin cannot be reached around the proxy. Optional bearer tokens are supported for a deployment that needs them, supplied out of band.

11 Deployment

11.1 Stack

Component	Detail
Host	Hetzner CPX32, Helsinki, Ubuntu LTS
Edge	Cloudflare proxy, SSL/TLS mode <i>Full (strict)</i>
Reverse proxy	Caddy with a Cloudflare Origin CA certificate; ACME disabled
API	uvicorn under systemd (robustuavs-api.service)
Client	Expo web export, output: "single", served statically
Bring-up	deploy/bootstrap.sh, deploy/cloud-init.yaml
Release	deploy/redeploy.sh then deploy/deploy.sh
Mail (separate)	deploy/mail/, its own compose project (§12)

Correction to v1. Version 1 described the reverse proxy as “Caddy, automatic TLS”. That was wrong. `deploy/Caddyfile` disables ACME and loads the certificate pair from `/etc/caddy/certs/`. The choice is deliberate: with an Origin CA certificate, nothing renews through the proxy and port 80 never has to be reachable from the whole internet for a challenge.

11.2 One command

```
ssh robustuavs 'cd /srv/robustuavs/repo && git checkout main \
&& ./deploy/redeploy.sh --watch'
```

`redeploy.sh` fetches, then runs `deploy.sh` detached under `setsid nohup`, so a dropped SSH session does not kill the build. Ctrl-C stops watching, not the deploy.

11.3 Three hard-won details

The build needs memory. `expo export` is the memory-hungry step, and `deploy.sh` refuses to run it below `MIN_BUILD_MB` (1500 MB) without at least 1 GB of swap. On a small instance it has previously taken the box down hard enough to lose `sshd`. The guard exists because that happened.

The origin cannot hairpin. A verification probe from the server to its own public hostname goes out through the CDN and returns 000, which looks like an outage when the site is fine. The probe pins the origin with `--resolve` instead.

Fetch, do not pull. `redeploy.sh` deliberately runs `git fetch` rather than `git pull`. `deploy.sh` already does `git reset --hard` plus `git clean -fd`, which recovers from any local mess on the deploy host. A pull cannot: it refuses to merge over a modified file and aborts the script, so the very reset that would have fixed the problem never runs. That once turned one dirty file into a permanently blocked deploy.

12 Project Mail

The platform has four official addresses, self-hosted on the same server so that no third party holds project correspondence and no forwarding service sits in the path. The stack lives in `deploy/mail/`, runs as its own compose project (`robustuavs-mail`), and is independent of `deploy.sh`: a platform redeploy never restarts mail, and a mail restart never touches the API.

Status. The mail package is built and verified offline, not installed. Its four scripts pass `bash -n`; the compose file passes `docker compose config`; the webmail virtual host, merged into the production Caddyfile, passes `caddy validate`; and the DKIM-record parser and account matcher were unit-tested. None of that proves a delivered message. Until the installer has run on the server and a test message to an external provider shows SPF, DKIM and DMARC all passing, the mailboxes below should be read as specified, not as live.

12.1 Architecture

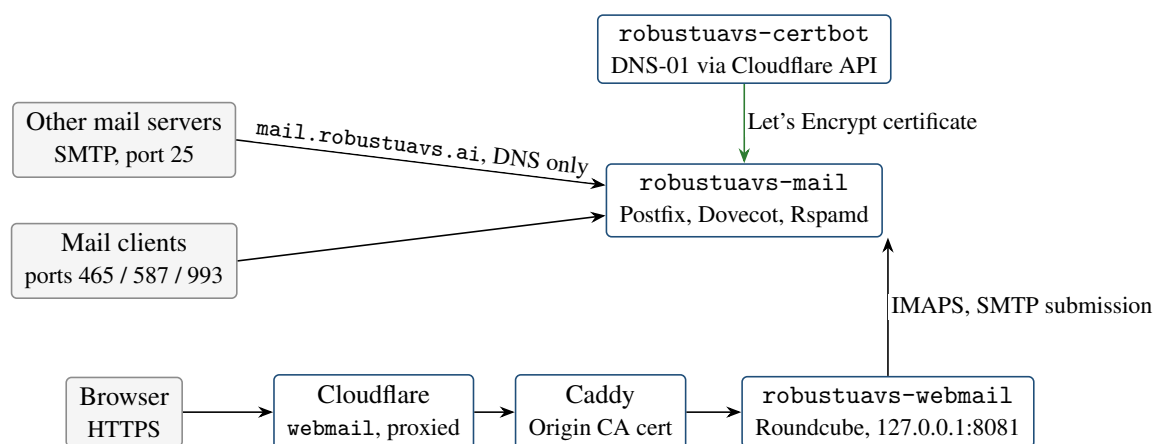


Figure 2: Mail and webmail paths. SMTP and IMAP reach the mail container directly; only the webmail page travels through Cloudflare and Caddy.

Component	Image (pinned)	Role
Mail server	<code>docker-mailserver:15.1.0</code>	Postfix (SMTP), Dovecot (IMAP), Rspamd (spam scoring, DKIM signing, SPF/DMARC checks), fail2ban, ManageSieve
Webmail	<code>roundcubemail:1.7.4-apache</code>	Browser mail at <code>https://webmail.robustuavs.ai</code> , published on loopback only
Certificates	<code>certbot/dns-cloudflare</code>	Let's Encrypt certificate for <code>mail.robustuavs.ai</code> by DNS-01 challenge; renewal checked every 12 h

Both service images are pinned to versions confirmed present in their registries. The mail server stays on the 15 series although 16 exists, because the provisioning sequence was verified against 15; the upgrade is a deliberate step rather than something inherited from a floating tag. Mail runs over IPv4 only, since an IPv6 address without a matching reverse-DNS record is rejected by major providers.

12.2 Mailboxes

Address	Purpose
roger@robustuavs.ai	Correspondence author. The address placed on submissions, so the one that most needs authentication to pass at strict institutional mail servers.
admin@robustuavs.ai	Operations. Receives postmaster@, abuse@, hostmaster@ and webmaster@ (RFC 2142 aliases), DMARC aggregate reports, and the certificate-authority registration.
support@robustuavs.ai	Public contact for the platform and the artifact.
noreply@robustuavs.ai	Reserved for anything the platform sends itself. Nothing uses it yet: the platform has no login or registration flow, deliberately, while the work is under double-blind review.

12.3 Two certificates, deliberately

The web platform and the mail server use different certificate authorities, and this is a requirement rather than an inconsistency. The web origin's Origin CA certificate is trusted only by Cloudflare, which is exactly its purpose. A mail client opening IMAPS on port 993, or another mail server negotiating STARTTLS on port 25, is not Cloudflare, and would reject it. The mail container therefore carries its own publicly trusted certificate, and the verification script fails loudly if an Origin CA certificate ever appears on port 993.

12.4 DNS records

Type	Name	Content	Proxy
A	mail	server IPv4	DNS only
A	webmail	server IPv4	Proxied
MX	@	mail.robustuavs.ai, priority 10	—
TXT	@	v=spf1 mx -all	—
TXT	mail._domainkey	v=DKIM1; k=rsa; p=... (2048-bit, selector mail)	—
TXT	_dmarc	v=DMARC1; p=quarantine; adkim=s; aspf=s; rua=...	—

SPF names mx rather than a on purpose. The apex record is proxied, so a would authorise Cloudflare's shared proxy range to send mail as the domain. mx authorises only the one host that actually sends. Reverse DNS for the server address must also be set to mail.robustuavs.ai at the hosting provider; no record in the zone can substitute for it.

 *The cost of the grey record. The deployment runbook keeps every web record proxied so the origin's address is never published, warning that one grey record exposes it permanently to passive-DNS archives. Mail breaks that rule unavoidably: the MX target must be reachable on port 25 by every sending server, and Cloudflare's proxy carries only HTTP(S). What is lost is concealment of the origin address; what is kept is the firewall rule restricting 80/443 to Cloudflare. The alternative, if concealment matters more, is to place mail on a second small server with its own address. `deploy/mail/README.md` sets out the trade and must be read before the record is saved.*

12.5 Operating it

```
sudo bash deploy/mail/setup-mail.sh --set-token    # Cloudflare token, hidden
input
sudo bash deploy/mail/setup-mail.sh --check        # verify everything, change
nothing
```

```

sudo bash deploy/mail/setup-mail.sh          # install; prints DNS
records + passwords once
sudo bash deploy/mail/install-caddy-vhost.sh  # webmail vhost, with backup
and rollback
bash deploy/mail/check-mail.sh               # DNS, TLS, ports, mailboxes
, outbound 25

sudo bash deploy/mail/mailuser.sh add|passwd|delete|alias|quota|list <user>

```

Three design decisions carry over from the RobustIDPS deployment this stack was adapted from. The installer verifies the Cloudflare token against the live API *before* contacting the certificate authority, because a bad credential otherwise burns issuance attempts and trips rate limits. It creates mailboxes *before* waiting for the server to report healthy, because the mail server refuses to start IMAP until at least one mailbox exists, so waiting first is a deadlock. And it never edits the live Caddyfile itself: adding the webmail host is a separate script that backs the file up, validates the result, and restores the backup if validation fails. Password resets are administrator-only by design: a reset link needs a trusted second channel, and for a mailbox the mailbox is that channel.

Two prerequisites sit outside the repository and no script can satisfy them: outbound port 25, which the host blocks by default on new projects (the installer tests it and reports rather than assuming), and inbound 25/465/587/993 on the provider’s cloud firewall if one is attached.

13 Search Visibility

SEO is embedded in the build rather than bolted on. The build runs `platform/mobile/scripts/inject-seo.mjs`, which injects per-route titles, descriptions, Open Graph and Twitter cards, JSON-LD (SoftwareApplication, Dataset, ScholarlyArticle) and Highwire citation tags at export time, with `robots.txt` and `sitemap.xml` shipped alongside. Yandex is covered as well as Google, since the work has a Russian institutional home.

■ *One decision is deliberately deferred. Author names in the JSON-LD and citation metadata are exactly what a double-blind venue must not be able to find from the artifact. Until the review status is settled, the choice is between stripping the author fields and adding `noindex`, and `docs/seo.md` records the trade-off rather than resolving it silently.*

14 Reproducibility

14.1 Ten minutes, no data download

```

git clone https://github.com/rogerpanel/RobustUAVs.ai && cd RobustUAVs.ai
python3 -m pip install jsonschema numpy scipy

python3 certificates/engine.py          # self-check, four certificate
constants
python3 experiments/make_paper_figures.py
python3 experiments/provenance_ledger.py
git diff --stat results/               # expect: no changes
python3 experiments/compose_certified.py

```

The `git diff` is the load-bearing step. The manuscript \inputs the generated blocks directly, so a non-empty diff means the paper and the data have drifted apart, and a reviewer can falsify the “machine-generated, not transcribed” claim in one command.

14.2 Full reproduction

`experiments/run_real_corpus.sh` runs `fetch`, `stage`, `ingest`, `validate`, `calibrate` and `figures`. It is idempotent and skips a stage whose inputs are absent, saying so. A missing dataset degrades the run; it

never silently fabricates it.

14.3 Anonymised artifact

`tools/anonymise.py` produces a de-identified copy for double-blind review and, more importantly, re-scans its own output against a token list defined independently of the rewrite rules, exiting non-zero on any survivor. On its first run it caught eleven real leaks the rules had missed. It deliberately does not strip `third_party/`, because that code is cited prior work rather than self-identification, and it does not touch binaries, listing them instead: PDFs carry author metadata that no text scan can see.

15 What Is Not Built, and What Is Blocked

15.1 Not built

WebSocket progress streaming (the client polls); ingestion as a background runner; the four steps in `docs/simulation_architecture.md` that would move the modelled fleet quantities to measured; the five platform additions of §16. Not yet *installed*, although built: the mail stack of §12.

15.2 Blocked on external access

Tracked in `PENDING_ON_DATA.md`, which is the complete list.

Item	What unblocks it
UAV-CAS stat CSV	The release file. Adapter is fixture-verified and the pipeline picks it up automatically.
Whelan attack intervals	Per-flight attack on/off times from the dataset documentation. The adapter refuses to fabricate them.
TEXBAT re-calibration	Real in-phase and quadrature recordings.
Platform diversity	Flights on a second and third airframe.
Flight regime	Cruise and transition segments.
PAC-Bayes numeric KL	<i>Not data-gated</i> . A computation over a committed checkpoint.
Anonymised artifact upload	Nothing external; run the scrubber and upload.

The two bold rows are the platform’s binding limitation. Everything else is downstream of the fact that no public capture observes both layers on one airframe with published attack intervals.

16 Research Programme

16.1 The parent paper

The platform is the artifact for *From Wire to Flight: A Conditional Mission-Completion Guarantee for UAVs*, targeted at IEEE S&P. Since v1 its abstract has been condensed from 455 to 283 words with every number and qualification retained (`paper/abstract_v7c_condensed.tex`), and it now states the measured interface rate as a supremum over the observed conditions rather than an average.

16.2 Five follow-on papers

v1 listed three follow-on directions. They have been reorganised, alongside two further lines of work, into five papers with distinct objects and named venues. Each attaches to a different link of the chain $\theta \mapsto \Delta(\theta) \mapsto \delta \mapsto \text{MCR}$, so they do not overlap. Proposals are in `paper/proposals/`, and `SUPERVISOR.BRIEF.tex` there is a two-page overview.

	Working title	Venue	Object
I	What Authenticity Costs in Metres	IEEE S&P	Authentication lowers γ and raises Δ ; a closed-form crossover decides which wins. Merges v1's directions 1 and 2.
II	SoK: The Quantifier Gap	IEEE S&P (SoK)	Guarantees and airworthiness objectives transfer only when their quantifiers match. Broadens v1's direction 3.
III	Certificates as Constraints	NeurIPS	A certified radius as a time-varying, pointwise constraint on a fleet allocation policy, feasible during learning.
IV	When the Attacker Is Indistinguishable from the Weather	USENIX Security	A source-agnostic perturbation budget spanning attacks, faults and environment, and a public cross-layer benchmark.
V	Keep Flying Whatever the Model Says	IEEE SaTML	A predictive safety filter that keeps a language-model planning agent inside the certified envelope whatever the agent outputs.

16.3 Platform work the programme implies

Every item below is **planned, not built**. Each reuses the existing export menu, provenance badge and certificate engine, and each will carry the same *measured / illustrative* discipline as the current pages.

	Platform addition	What it does
I	<i>Certified Budget</i> page; /api/certify/budget	Choose sensors and their authentication, a transport, a signature scheme, a message rate and a margin; see the remaining budget, the crossover sign and bus occupancy live. Replaces v1's separate PNT-selector and crypto-budget ideas.
II	<i>Compliance Pack</i> export	One action emitting a SORA Annex A-shaped evidence bundle for a chosen operating point, with every assumption listed as an open obligation rather than hidden in the result.
III	Certified fleet allocation	A constrained allocation layer over the fleet simulator, with the certified radius as a hard constraint; requires raising the current MAX_FLEET = 12 limit.
IV	Fault and environment terms; benchmark release	Two new budget terms in the certificate engine, and a public release of the cross-layer benchmark with enforced provenance flags.
V	<i>Safety Filter</i> page	A language-model agent in the mission-planning loop, behind a predictive safety filter; shows the agent's proposal, the filtered action, and the fallback rate under adaptive evaluation.

■ *What the programme does not change. None of the five papers removes the binding limitation in §15.2: the interface rate is still calibrated in hover on one airframe, and no public capture observes both layers on one platform with published attack intervals. Paper I adds navigation sources and Paper IV adds perturbation sources, but neither is a cruise campaign on a second airframe.*

17 Conclusion

RobustUAVs.ai is a working instrument for a question that previously had no instrument: whether a mission completes when an attacker evades the detector by construction. It unifies six corpora into one representation whose unit is the cross-layer pairing, implements four certificates and the composition that chains them, and serves the result through 25 pages that badge every number with its provenance.

The platform's distinguishing property is not any single capability but its posture. It reports the number of strongest-evidence pairings as zero. It shows a Calibration page that computes nothing and explains why. It uses the larger of two Lipschitz constants because the smaller one would flatter the

result. It caught, and documents, a measurement artifact that would have produced a spectacular but wrong finding. A research platform earns trust by being checkable, and being checkable means making the gaps as visible as the results.

Version 2 leaves that core untouched and builds around it: four project addresses with authenticated mail, a corrected account of how the origin terminates TLS, and a research programme in which each of five papers names the platform work it needs. The two flight-campaign items that remain genuinely blocked are still the limit on what any of it can claim, and the next version is measured against them.

Software Availability

Platform	https://robustuavs.ai
Artifact	https://github.com/rogerpanel/RobustUAVs.ai
Corpus	Kaggle, DOI 10.34740/kaggle/dsv/18346203
Licence	MIT. Third-party data stays under its own licence; none is redistributed.
Contact	support@robustuavs.ai (<i>once the mail stack is installed</i>)

Documents worth opening, in order of usefulness to a reviewer:

1. docs/certified_regime_analysis.md — the analysis that decided the headline framing, including a refuted hypothesis kept in the record;
2. docs/interface_campaign_results.md;
3. docs/SCHEMA.md;
4. results/paper_figures/README.md — the generated provenance ledger;
5. PENDING_ON_DATA.md;
6. new in v2: deploy/mail/README.md and paper/proposals/README.md.